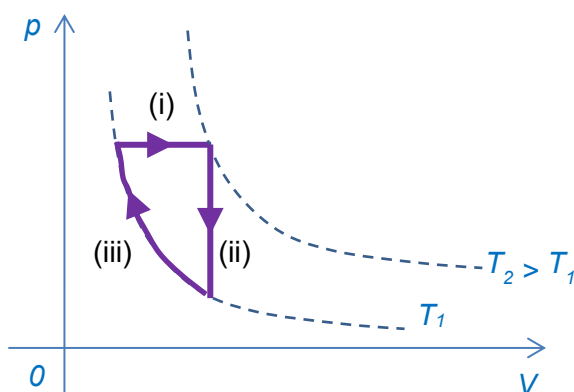


15	An ideal gas undergoes a cyclic process in three stages: (1) Expansion at constant pressure, (2) Decrease in pressure at constant volume, (3) Compression at constant temperature. Which of the following must be true?
A	There is net heat released by the gas during the cycle.
B	The net work done on the gas during the cycle is positive.
C	The net work done by the gas during the cycle is equal to the net heat transferred to the gas.
D	The change in internal energy of the gas is equal to the work done during the constant pressure stage minus the work done during the constant temperature stage.

Answer: C

Represent the cyclic process on a p-V diagram:



From the p-V diagram there is **NET** work done **BY** gas during the cycle, so net **W** is negative
→ Eliminate option B.

Since net work done is non-zero, Option D would imply that ΔU is non-zero but in fact per cycle ΔU is zero.

Per cycle, $\Delta U = 0$
→ Eliminate option D.

By $\Delta U = Q + W$, $0 = Q + (\text{negative value})$, so net **Q** is positive which implies that there is net heat transferred **TO** the gas during the cycle
→ Eliminate option A.

Conversely, option C is correct.

16 A beam of plane polarised light of intensity I falls normally onto a thin sheet of polaroid.